

Strongly Nondegenerate Forms and their Associated Structures on Higher Knot Spaces

Given a Riemannian manifold M with special holonomy, one obtains a distinguished parallel differential form on it, called a strongly nondegenerate form. In this talk, we show how such a form gives rise to an almost Kaehler structure on the space of higher-dimensional knots in M , and how its (infinite-dimensional) symplectic geometry corresponds to the calibrated geometry of M . We will then discuss the special case when M is Calabi-Yau, in which we obtain additional holomorphic analogues of these results.

Section 1: Vector Cross Products and Strongly Nondegenerate Forms

Definition

(M^n, g) Riemannian manifold. A vector cross product of degree r is a smooth bundle map:

$$\chi : \Lambda^r TM \rightarrow TM$$

satisfying $g(\chi(v_1, \dots, v_r), v_j) = 0$ and $g(\chi(v_1, \dots, v_r), v_{r+1}) = |v_1 \wedge \dots \wedge v_r|^2$. This induces an $r+1$ form, called a vector cross product form or strongly nondegenerate form via $\eta(v_1, \dots, v_{r+1}) = g(\chi(v_1, \dots, v_r), v_{r+1})$, which is characterized by its degree, n , and the fact that $(v_1 \wedge \dots \wedge v_r) \lrcorner \phi = 0$ if and only if $v_1, \dots, v_r$ are linearly dependent.

Examples

$r = 1$ this is a Hermitian complex structure and its SND form is the Kaehler form.

$r = n - 1$ its SND form is the volume form.

$r = 2, n = 7$ is the associative 3-form of a G_2 structure.

$r = 3, n = 8$ is the Cayley 4-form.

Note that the existence of these are equivalent to a reduction of the structure group of the frame bundle from $O(n)$ to $U(m)$, $SO(n)$, G_2 , $Spin(7)$ respectively.

Theorem

These are the only possible vector cross products (in finite dimensions).

Sketch:

Let $V_{r,n}$ denote the Stiefel manifold of orthonormal tuples of r vectors in $\mathbb{R}^n$. The map

$$(f_1, \dots, f_{r-1}) \mapsto (f_1, \dots, f_{r-1}, \frac{\chi(f_1, \dots, f_{r-1})}{|\chi(f_1, \dots, f_{r-1})|})$$

is a section of the projection $V_{r,n} \rightarrow V_{r-1,n}$. By algebraic topology (due to Eckmann, Whitehead 1963), such sections only exist in the cases when $r = 1, r = n - 1, (r, n) = (2, 7), (r, n) = (3, 8)$. ■

We now look at the natural objects that VCP and SND forms interact with: Fix (M^n, g, χ, ϕ) with ϕ closed.

Def

An $r+1$ dim submanifold A is an instanton if it is preserved by χ .

Remark

Instantons are calibrated by ϕ , in particular, minimal submanifolds.

Examples

$r = 1$, holomorphic curves in Kahler manifolds.

$r = n - 1$, open domains.

$r = 2$, associative submanifolds.

$r = 3$, Cayley submanifolds.

Def

A submanifold C is called a brane if $\phi|_C = 0$ and $\dim C$ is maximal (which equals $(n + r - 1)/2$).

Examples

- $r = 1$ Lagrangian submanifolds.
- $r = n - 1$ hypersurfaces.
- $r = 2$ coassociative submanifolds.
- $r = 3$ do not exist.

Manifolds M ($\dim M$)	VCP form ϕ (degree of ϕ)	Instanton A ($\dim A$)	Brane C ($\dim C$)
Oriented mfd. (n)	Volume form (n)	Open Subset (n)	Hypersurface ($n - 1$)
Kähler mfd. ($2m$)	Kähler form (2)	Holomorphic Curve (2)	Lagrangian Submanifold (m)
G_2 -manifold (7)	G_2 -form (3)	Associative Submanifold (3)	Coassociative Submanifold (4)
$Spin(7)$ -mfd. (8)	Cayley form (4)	Cayley submfd. (4)	N/A

Induced symplectic geometry on knot spaces

Let Σ be an oriented closed $r - 1$ -submanifold of M . Consider $Emb(\Sigma)$, the set of embeddings $\Sigma \rightarrow M$ and $ev : \Sigma \times Emb(\Sigma) \rightarrow M$ the evaluation map. There is a right action of $Diff(\Sigma)$ on $Emb(\Sigma)$ by composition.

We define a 2-form on $Emb(\Sigma)$ by $\omega = \int_{\Sigma} ev^* \phi$, which explicitly for u, v vector fields (note that a tangent vector to $Emb(\Sigma)$ at f is an element of $\Gamma(\Sigma, f^*(TM))$) is given by $\omega(u, v) = \int_{\Sigma} (u \wedge v) \lrcorner \phi$. This isn't a symplectic 2-form, because $((u \wedge v) \lrcorner \phi)|_{\Sigma}$ is not top-degree if u is tangent to Σ , so ω is degenerate along tangent directions to the orbits of the action of $Diff(\Sigma)$ on $Emb(\Sigma)$, so if we take $Emb(\Sigma)/Diff(\Sigma)$, ω descends to a well-defined 2-form. We denote $Emb(\Sigma)/Diff(\Sigma) =: \mathcal{K}$ (for knot space). Note that tangent vectors of $\mathcal{K}$ are sections of the normal bundle of Σ in M .

ω is closed since ϕ is closed:

$$d\omega = d \int_{\Sigma} ev^* \phi = \int_{\Sigma} ev^* d\phi = 0$$

Remark

bring up $\Sigma = S^1, M = \mathbb{R}^3$ as the classical knot space, hence the name

Now, we can make this into an almost Kaehler manifold as follows:
we have a natural L^2 metric on $\mathcal{K}$ given by

$$g_{[f]}^{\mathcal{K}}(u, v) = \int_{\Sigma} g(u, v) \nu_{\Sigma}$$

where ν_{Σ} is the induced volume form on Σ . Define an almost complex structure J on $\mathcal{K}$ in the usual way,

$$\omega_{[f]}(u, v) = g_{[f]}^{\mathcal{K}}(J_{[f]}(u), v)$$

Prop

This J is a Hermitian almost complex structure.

Proof:

It suffices to prove that J is a VCP for $r = 1$. This follows since from expanding the definitions, one can conclude the action of J is given by $J(u) = \chi(e_1, \dots, e_{r-1}, u)$ (as a map in $\text{End}(\Gamma(\Sigma, N_{\Sigma/M}))$), where $(e_1, \dots, e_{r-1})$ is an oriented orthonormal frame for Σ . But $\chi(e_1, \dots, e_{r-1}, \cdot)$ gives a 1-fold VCP. ■

Thus $\mathcal{K}$ is an almost Kahler manifold.

Thus, we have symplectic geometry on $\mathcal{K}$ induced by the VCP geometry on M . This suggests “the localization principle”, which says that geometry of ϕ should correspond to symplectic geometry of M . The first instance of this is the converse to ϕ closed implies $\mathcal{K}$ almost Kahler:

Lemma (Localization)

A form η of degree $k > r - 1$ on M vanishes if the $(k - r + 1)$ form on $\mathcal{K}$ obtained by transgression vanishes.

Proof:

We must show $\eta_p(v_1, \dots, v_k) = 0$ for $p \in M, v_1, \dots, v_k \in T_p M$. wlog assume $v_1, \dots, v_k$ orthonormal. then choose an embedding $f \in \text{Emb}(\Sigma)$ with $p \in f(\Sigma)$ s.t. $v_1, \dots, v_{r-1}$ are tangential, $v_r, \dots, v_k$ are normal. choose sections $\hat{v}_r, \dots, \hat{v}_k \in \Gamma(\Sigma, f^*(TM))$ which at p agree with $v_r, \dots, v_k$. By multiplying v_r with a sequence of functions on Σ approaching the dirac delta at p , we obtain sections $(\tilde{v}_r)_\varepsilon$ which approach $\delta(p)v_r$ as $\varepsilon \rightarrow 0$, and so

$$\eta_p(v_1, \dots, v_k) = \lim_{\varepsilon \rightarrow 0} \left(\int_{\Sigma} ev^* \eta \right) ((\tilde{v}_r)_\varepsilon, \hat{v}_{r+1}, \dots, \hat{v}_k)$$

since we assume $\int_{\Sigma} ev^* \eta = 0$, we obtain $\eta = 0$. ■

Corollary

$(\mathcal{K}, \omega)$ is almost Kaehler iff ϕ is closed.

basically, the above argument says that in the higher knot space context, we can reduce many analytic conditions to analytic conditions on $M + \text{linear algebra}$ at a point.

Example

Suppose v is a vector field on M preserving ϕ . Then it lifts to a vector field V on $\mathcal{K}$ preserving ω , and moreover, if v is ϕ -Hamiltonian, i.e. $v \lrcorner \phi = d\eta$ for some η , then we can define $F \in C^\infty(\mathcal{K})$ by $F(f) = \int_{\Sigma} f^* \eta$ which is ω -Hamiltonian for V .

Calibrated geometry

In the infinite dimensional context, a submanifold L of $\mathcal{K}$ is Lagrangian if L is maximal on which ω vanishes, iff ω vanishes and any vector field ω -orthogonal to L is tangent along L .

Notation

For a submanifold $C \subseteq M$, denote $\mathcal{K}(C)$ the same construction as above but applied to embeddings $\Sigma \hookrightarrow C$.

Theorem

$C \subseteq M$ is a brane iff $\mathcal{K}(C)$ is a Lagrangian submanifold of $\mathcal{K}$.

Proof:

First, suppose $\mathcal{K}(C)$ is Lagrangian. We need to show ϕ vanishes on C and C is of the appropriate dimension. the maximal dimension of Lagrangian-ness corresponds to maximal dimension of C provided ϕ does indeed vanish (by localization arguments). For this part, by Lagrangian-ness,

$$0 = \omega_{[f]}(u, v) = \int_{\Sigma} (u \wedge v) \lrcorner (ev^* \phi)$$

for $[f] \in \mathcal{K}(C)$ for any u, v tangent vectors. by applying the localization argument, we then obtain that for any $x \in \Sigma$, $\phi(u_x, v_x, e_1, \dots, e_{r-1}) = 0$ where $e_1, \dots, e_{r-1}$ are oriented orthonormal vectors in $T_x \Sigma$. vary u, v , and f to cover TC , which shows $\phi|_C = 0$.

Conversely, $\phi|_C = 0$ implies $\omega|_{\mathcal{K}(C)} = 0$, so we need to show $\mathcal{K}(C)$ is maximal dimension if C is a brane. Thus suppose there is a section $v \in \Gamma(\Sigma, N_{\Sigma/M})$ but not in $\Gamma(\Sigma, N_{\Sigma/C})$. Then by localization arguments, for any $x \in f(\Sigma)$, ϕ vanishes on the linear subspace of $T_x M$ spanned by v_x and $T_x C$, contradicting maximality of C . ■

The same sort of argument works for instantons. First, what are the instantons in $\mathcal{K}$:

Def

Let D^2 be a disk in $Emb(\Sigma)$ such that for all $v \in T_{[f]} D$, the corresponding vector field in $\Gamma(\Sigma, f^*(TM))$ (under the preimage of the quotient) is normal to Σ . We call D a normal disk. From this, we can form the $r + 1$ -submanifold

$$A := \bigcup_{f \in D} f(\Sigma) \subseteq M$$

which is an embedding if D is small enough (this is diffeomorphic to $D \times \Sigma$). This gives a disk in $\mathcal{K}$ denoted $\hat{D} := \pi(D)$, where π is the quotient map (one can also phrase this in terms of a connection over the principal fibration $Diff(\Sigma) \rightarrow Emb(\Sigma) \rightarrow^\pi \mathcal{K}$).

Theorem

For a normal disk D in $Emb(\Sigma)$, $\hat{D}$ is a J -holomorphic disk in $\mathcal{K}$, that is, calibrated by ω , if and only if A is an instanton in M and $A \rightarrow D$ is a Riemannian submersion.

Proof:

First, assume $\hat{D}$ is a J -holomorphic disk. Let $[f] \in \hat{D}$, and $u, v \in T_{[f]}(\hat{D}) \subseteq \Gamma(\Sigma, N_{\Sigma/A})$. By calibration, we have $\phi(u, v, e_1, \dots, e_{r-1}) \leq |u \wedge v|$ for $e_1, \dots, e_{r-1}$ oriented orthonormal frame on $f(\Sigma)$. Thus integrating and using the fact ω corresponds to a 1-fold VCP w.r.t the L^2 metric, we have

$$\int_{f(\Sigma)} |u \wedge v| \geq \int_{f(\Sigma)} (u \wedge v) \lrcorner (ev^* \phi) \stackrel{J-hol}{=} \sqrt{|u|_{\mathcal{K}}^2 |v|_{\mathcal{K}}^2 - \langle u, v \rangle_{\mathcal{K}}^2} = \left(\int_{f(\Sigma)} |u|^2 \int_{f(\Sigma)} |v|^2 - \left(\int_{\Sigma} g(u, v) \right)^2 \right)^{1/2}$$

and Hölder's inequality shows we have an equality. The equality $\int_{f(\Sigma)} |u \wedge v| = \int_{f(\Sigma)} (u \wedge v) \lrcorner (ev^* \phi)$ says A is an instanton, and the equality

$$\left(\int_{f(\Sigma)} |u \wedge v| \right)^2 + \left(\int_{\Sigma} g(u, v) \right)^2 = \int_{f(\Sigma)} |u|^2 \int_{f(\Sigma)} |v|^2$$

is equivalent to the submersion condition (it says for any $[f]$, there are constants C_1, C_2 such that $|u_x| = C_1 |v_x|$ and the angle between u_x, v_x is C_2). all these steps are reversible. ■

Remark

With similar techniques, one can consider the symplectic knot spaces, which are the symplectic embeddings of a symplectic submanifold Σ in a symplectic manifold M , on which one can prove similar correspondences about isotropic, coisotropic, lagrangian submanifolds, tame/compatible curves etc.

Holomorphic analogue

We will now get a holomorphic analogue of the above results. First, we need a notion of complex vector cross product:

Definition

On a Kahler manifold of complex dimension n , an r complex VCP is $\phi \in \Omega^{r+1,0}(M)$ such that $|(e_1 \wedge \dots \wedge e_r) \lrcorner \phi| = 2^{(r+1)/2}$ for orthonormal $(1, 0)$ tangent vectors $e_1, \dots, e_r$.

Lemma

The only complex vector cross products are $r = n - 1$ (holomorphic volume form) and $r = 1$ (holomorphic

symplectic form), corresponding to Calabi-Yau and hyperkaehler (for the latter, $\phi = \omega_I - i\omega_K$.)

Definition

An $r + 1$ dimensional submanifold $A \subseteq M$ is an instanton with phase θ if $\text{Re}(e^{i\theta}\phi)$ calibrates it.

For boundary conditions, branes, we now obtain two options:

Definition

A Neumann brane (N-brane) is a $n + r - 1$ -dimensional submanifold C with $\phi|_C = 0$, and a Dirichlet brane (D-brane) with phase θ if $\omega|_C = 0, \text{Re}(e^{i\theta}\phi)|_C = 0$.

Example

Instantons of a Calabi-Yau are special lagrangian with phase θ , with N-branes complex hypersurfaces and D-branes a SLAG with phase $\pi - \theta/2$.

Instantons of a hyperkaehler manifold of phase θ are J_θ -holomorphic curves, where $J_\theta = \cos \theta I + \sin \theta K$. D-branes are $J_{\theta+\pi/2}$ complex Lagrangian, and N-branes are J -complex Lagrangian. Taking $\theta = 0$, we obtain instantons are I -complex Lagrangian, N-branes J -complex Lagrangian, D-branes K -complex Lagrangian.

We would like a similar localization principle on $\mathcal{K} := \text{Emb}(\Sigma)/\text{Diff}(\Sigma)$ as before (note the symplectic reduction). We would expect the holomorphic volume form to give as before a holomorphic symplectic form, to make $\text{Emb}(\Sigma)/\text{Diff}(\Sigma)$ a hyperKaehler knot space.

Problem 1: need to fix a background volume form on Σ to construct a symplectic structure, and this is only preserved by the volume-preserving diffeomorphisms on Σ .

Solution: Just take the quotient by the volume-preserving diffeo's.

Problem 2: This quotient space is not hyperkaehler, since the form is degenerate.

Solution:

do more reduction to kill the degeneracy.

Problem 3: this is still not hyperkaehler, since the kaehler form isn't in general closed.

solution: doesn't matter that much.

Fix a CY n -fold M with a holomorphic volume form Ω_M , and Σ an $n - 2$ dimensional submanifold Σ without boundary with $b^1(\Sigma) = 0$. Fix a volume form σ on Σ . Then the Kaehler form ω gives a symplectic form on $\text{Emb}(\Sigma)$ as follows:

$$\omega_e(X, Y) = \int_{\Sigma} \omega(X, Y) \sigma$$

This is preserved by volume-preserving diffeos of (Σ, σ) . This action is Hamiltonian on the set of $f \in \text{Emb}(\Sigma)$ s.t. $f^*([\omega]) = 0 \in H^2$ with moment map μ given by $f \mapsto \alpha$, where α is any 1-form satisfying $d\alpha = f^*\omega$ (this is a map from the above set to Ω^1 minus exact forms, which is naturally identified with the dual of the Lie algebra of $\text{Diff}(\Sigma, \sigma)$). In particular $\mu^{-1}(0)$ is the set of isotropic embeddings of Σ into M , so we can form the symplectic quotient $\text{Emb}(\Sigma)/\text{Diff}(\Sigma, \sigma)$, which is close to the moduli space of isotropic submanifolds in M (which is where we take the stronger quotient by $\text{Diff}(\Sigma)$).

Remark

talk about how this is basically isotropic submanifolds/ $\text{Diff}(\Sigma) \otimes \mathbb{C}$.

Remark: The isotropic knot space $\hat{\mathcal{K}}_\Sigma M$ is a quotient space of $\mu^{-1}(0)/\text{Diff}(\Sigma)$, the space of isotropic submanifolds in M . If $f : \Sigma \rightarrow M$ parametrizes an isotropic submanifold in M , then deforming Σ along T_Σ directions simply changes the parametrization of the submanifold $f(\Sigma)$, namely the equivalence class in $\mu^{-1}(0)/\text{Diff}(\Sigma)$ remains unchanged. However if we deform Σ along $T_\Sigma \otimes \mathbb{C}$ directions, then their equivalence classes in $\hat{\mathcal{K}}_\Sigma M$ remains constant. Notice that the isotropic condition on Σ implies that $T_\Sigma \otimes \mathbb{C} \cong T_\Sigma \oplus J(T_\Sigma) \subset f^*T_M$. Therefore, roughly speaking, $\hat{\mathcal{K}}_\Sigma M$ is the space of isotropic submanifolds in M divided by $\text{Diff}(\Sigma) \otimes \mathbb{C}$. In particular, the tangent space of $\hat{\mathcal{K}}_\Sigma M$ is given by

$$T_{[f]}(\hat{\mathcal{K}}_\Sigma M) = \Gamma(\Sigma, f^*(TM)/(T_\Sigma + JT_\Sigma))$$

We now use Ω_M to get a holomorphic symplectic structure on (a quotient of) this quotient. Define a complex closed 2-form $\Omega_E = \int_\Sigma ev^* \Omega_M = \omega_I - i\omega_K$, where $\omega_I = \int_\Sigma ev^* \text{Re}(\Omega_M)$ and $\omega_K = -\int_\Sigma ev^* \text{Im}(\Omega_M)$, with corresponding endomorphisms I and K on $T_f(\mu^{-1}(0))$ in the usual Kaehler way (via the L^2 metric), $\omega_I(A, B) = g(IA, B)$ etc. We will form the isotropic knot space as the symplectic reduction of $\mu^{-1}(0)$ by ω_I or ω_K , and show it has a holomorphic symplectic structure. To do this, we define a distribution $\mathcal{D}$ on $\mu^{-1}(0)$ by $\mathcal{D}_f = \{X \in T_f(\mu^{-1}(0)) : X \lrcorner \omega_{I,f} = 0\}$ (this is the “degenerate bit”).

Lemma

$$\mathcal{D}_f = \Gamma(\Sigma, T\Sigma + JT\Sigma)$$

proof:

expand definitions, use the fact that $\mu^{-1}(0)$ are isotropic embeddings, and apply localization.

The rank of the subbundle $T\Sigma + JT\Sigma$ in $f^*(TM)$ is $2(n-2)$ (so constant rank), so since it's the isotropic bit we need to quotient out by, coisotropic reduction says we can take the symplectic reduction $\hat{\mathcal{K}} := \mu^{-1}(0)/\langle D \rangle$, where we take the equivalence relation generated by the leaves of the foliation (since $\mathcal{D}$ is integrable because ω_I is closed).

Its tangent spaces are given by

$$T_{[f]}(\hat{\mathcal{K}}) = \Gamma(\Sigma, f^*(TM)/(T\Sigma + JT\Sigma))$$

We will now show $\hat{\mathcal{K}}$ is a (integrable) Hermitian complex manifold with a holomorphic symplectic form Ω (descended from Ω_E), (i.e. a 1-fold complex VCP), but this is not in general hyperkaehler because the kaehler form might not be closed (the descendent of ω_e). However, there are three almost complex structures on $\hat{\mathcal{K}}$, $\hat{I}, \hat{J}, \hat{K}$, which obey the quaternion relations and the associated Kaehler forms $\hat{\omega}_I, \hat{\omega}_K$ are closed, which implies $\hat{J}$ is integrable. All that is left is $\hat{\omega}_J$ (descendent of ω_e) being closed to be hyperkaehler.

Lemma

$$IJ = -JI = K, JK = -KJ = I \text{ on } \mu^{-1}(0).$$

proof:

straightforward but long algebra after expanding definitions in terms of integrals of things on M . ■

Theorem

With the above setup, $\hat{\mathcal{K}}$ is integrable (so Hermitian complex).

Proof:

since $\hat{\omega}_I, \hat{\omega}_K$ are closed, a result of Hitchin shows that if the quaternion relations hold for $\hat{I}, \hat{J}, \hat{K}$ then $\hat{J}$ is integrable. By the previous lemma, one only needs to check $\hat{I}^2 = \hat{K}^2 = -\text{id}$. By localization arguments, we reduce this to proving it linear algebraically at a point, at which the closure of $\hat{\omega}_J$ means nothing, so we have a linear hyperkahler structure at a point, which means the quaternion relations hold, as desired. ■

Remark

If M is a CY 3-fold, then $b_1(\Sigma) \neq 0$ (it's a circle), so we need to modify this construction. this gives

a symplectic structure on a rank $b_1(\Sigma)$ torus bundle gerbe, which is again approximately the space of isotropic submanifolds of M .

we will now try to illustrate the localization principle for $\widehat{\mathcal{K}}$:

Proposition

Let D be a normal disk in $\mu^{-1}(0)$ with A defined as before (after taking the appropriate quotient), and assume $A \rightarrow D$ is a Riemannian submersion. Then the reduction $\widehat{D}$ in $\widehat{\mathcal{K}}$ is an instanton with phase θ ($\widehat{J}_\theta$ holomorphic curve) iff A is an instanton in M (SLag phase θ).

proof:

replace ϕ with $\text{Re}(e^{i\theta}\phi)$ and apply previous proposition (one needs to check $\widehat{D}$ remains a disk in $\widehat{\mathcal{K}}$, which is straightforward from definitions). ■

We conclude with two theorems that summarize the same principle:

First, for a submanifold C , $\widehat{\mathcal{K}}(C)$ denotes $\{Emb(\Sigma, C) \cap \mu^{-1}(0)\}/\langle \mathcal{D} \rangle$. These theorems are proved with similar techniques localization techniques to previously, though with more technical details when talking about the relevant (very many) complex structures involved.

Theorem

Let C be a connected analytic (probably unnecessary) submanifold of M . Then C is an N-brane in M (i.e. complex hypersurface) iff $\widehat{\mathcal{K}}(C)$ is a $\widehat{J}$ -complex Lagrangian submanifold in $\widehat{\mathcal{K}}$.

Theorem

Let L be a connected analytic submanifold of M . Then L is a D-brane with phase θ in M (i.e. SLag with phase $\theta - \pi/2$) iff $\widehat{\mathcal{K}}(L)$ is a $\widehat{J}_{\theta+\pi/2}$ -complex Lagrangian submanifold in $\widehat{\mathcal{K}}$.

future efforts: make some version of this principle work to obtain a quaternion/g2/spin(7) analogue of this holomorphic analogue.